Computers and Education Open 11 (2026) 100398

Contents lists available at ScienceDirect

Computers and Education Open

journal homepage: www.sciencedirect.com/journal/computers-and-education-open

Temporal dynamics of online learning interactions: A learning analytics study within the CoI framework

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ARTICLE INFO

Keywords:
Digital literacy
Older adults
Peer support
Community of Inquiry (CoI) framework
Online learning

ABSTRACT

Understanding how online learning unfolds over short time spans is essential for improving the design and facilitation of digital learning environments. This study examines the temporal dynamics of online learning interactions through the Community of Inquiry (CoI) framework in two two-week online courses for older adults. Using Moodle log data from an instructor-led edition ($n = 27$) and a peer-facilitated edition ($n = 45$), 4,276 behavioural traces were mapped to teaching, social, and cognitive presence and transformed into daily time series. Cross-correlation analyses ($\pm$3-day lag) were conducted to explore short-term temporal associations. The patterns observed suggest that synchronous videoconferencing may precede increases in cognitive engagement, and that learning guides may be followed by brief sequences of individual exploration and forum participation. Peer facilitation appears to be associated with higher levels of social presence, although temporal coupling patterns across presences remain broadly comparable. While exploratory in nature, these findings offer preliminary indications of short-term temporal structures in online engagement and may inform the timing of instructional scaffolds and facilitation strategies in courses designed for older adults.

1. Introduction

Learning Management Systems (LMSs) automatically record extensive, time-stamped data on participants' interactions, including resource access, activity completion, forum contributions, and connection patterns. These behavioural traces, while limited to observable actions, offer a scalable and replicable means of examining how learning unfolds over time in digital environments. Prior research has shown that such metrics can reveal how learning unfolds in digital environments and inform the personalisation and improvement of educational provision [1,2]. Empirical work has demonstrated the utility of such metrics in explaining academic performance [3–6], learner satisfaction and technology acceptance [7,8], and early withdrawal from courses.

Log-based analyses have also been used to explore learner autonomy, interaction patterns, and collaborative processes [9–14]. Recent studies further link behavioural traces to dimensions of engagement—social, cognitive and behavioural—highlighting their analytical value for learning analytics [15–18]. However, despite the growing use of LMS data, relatively few studies examine the *temporal* structure of online engagement, particularly in short-duration courses where interaction rhythms may be especially salient. This gap is especially relevant for understanding how specific learning activities or facilitation strategies trigger subsequent behaviours, and how these patterns unfold among older adult learners—an increasingly important population in digital education.

This study addresses this gap by examining temporal relationships among behavioural indicators derived from LMS logs in two online courses for older adults. Using a time-series approach, we analyse how teaching, social, and cognitive presence—operationalised through log-based proxies—interact over short time lags, and how these dynamics differ across two facilitation structures: an instructor-led edition and a peer-supported edition. The study contributes to research on the Community of Inquiry (CoI) framework, process-oriented learning analytics, and the understanding of engagement rhythms among older adults.

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https://doi.org/10.1016/j.caeo.2026.100398
Received 30 December 2025; Received in revised form 29 May 2026; Accepted 18 July 2026
Available online 18 July 2026

2. Theoretical background: the modified CoI framework

The CoI framework conceptualises online learning as the interplay of social, cognitive, and teaching presence (Garrison, Anderson & Archer, 2000, [19–21]). Social presence refers to open communication and interpersonal connection; cognitive presence describes the phases of meaning-making; and teaching presence encompasses design, facilitation, and instructional guidance.

Analyses of LMS logs have been used to operationalise these constructs by linking observable behaviours to pedagogical processes [1]. Studies show that behavioural patterns relate to performance, motivation, and self-regulation [22]. Research has also shown that student-content interactions often underpin cognitive presence, while peer and instructor exchanges support social and teaching presence [23]. Forum analyses further indicate that teaching presence can scaffold social presence, which in turn supports cognitive engagement [24, 25]. Recent research in large-scale online courses has also highlighted the temporal nature of cognitive engagement. Liu et al. [26] showed that emotional and cognitive engagement in MOOC discussion forums evolve dynamically over time and jointly predict learning achievement. Their findings reinforce the importance of examining short-term temporal patterns in online learning and support the use of log-based indicators to capture fluctuations in cognitive presence.

Shea and Bidjerano [27] proposed extending the CoI framework with learning presence, conceptualised as self-regulated learning (SRL), to account for how planning, monitoring, and regulation mediate relationships among the presences [28–31]. Integrating SRL into the CoI framework broadens its explanatory power [32–38">]. In this study, SRL is used as a *conceptual lens* to interpret how learners may organise their activity sequences, although SRL is not directly measured.

Evidence suggests that teaching presence often initiates interaction cycles, strengthening social presence and stimulating cognitive presence (Garrison et al., 2000; [24]). From an SRL perspective, access to guides and tasks may trigger planning and monitoring processes that structure subsequent engagement [11,21,27,39,40]. Learning guides, in particular, have been shown to prompt exploration and collaborative engagement [41,42].

Rather than a fixed sequence, knowledge construction often follows reciprocal loops between individual and collaborative activity, shaped by instructional context and learners' regulatory processes ([43]; Pardo et al., 2019). Temporal analyses can help reveal these loops by identifying short-term behavioural couplings across different types of activity.

3. Contextual factors relevant to this study

Although the present study does not compare older adults with younger populations, prior research highlights distinctive patterns in ageing and digital learning. Older adults often rely more on structured guidance, progress through tasks more sequentially, and show greater sensitivity to cognitive load and interface complexity ([44]; Knowles et al., 2015). These characteristics may shape how they engage with online resources and how they respond to instructional scaffolding.

Research on peer-supported learning also suggests that facilitation structures influence participation rhythms and perceptions of social support [45]. Analysing two course editions—one instructor-led and one peer-facilitated—provides an opportunity to observe how different forms of support may shape temporal engagement patterns, without aiming to compare their effectiveness.

These contextual considerations complement the CoI framework and support the relevance of examining temporal dynamics in online learning among older adults.

4. Research questions

Drawing on the preceding theoretical background, this study addresses three interrelated research questions about the temporal relations among teaching, cognitive and social presence, operationalised via log-based proxies of CoI-related activity.

Rationale: Teaching presence and cognitive/social presence. We examine whether synchronous teaching presence (attendance at videoconferencing sessions) scaffolds planning and clarification, producing subsequent increases in resource access and forum interaction.
RQ1. What temporal relationship exists between access to videoconferencing (as an indicator of teaching presence) and actions associated with cognitive presence (access to learning guides, content, and assignments) and social presence (forum participation)?
Rationale: Cognitive as organiser of task cycles. We test whether access to learning guides and assignments, as proxies for cognitive presence, structure task cycles that alternate individual study and collaborative discussion when students seek clarification or share findings.
RQ2. What temporal relationship exists between access to the learning guide and assignments—as elements that activate cognitive and social presence—and the exploration of content and participation in forums?
Rationale: Dynamics across course phases. We investigate how the cognitive–social relationship varies over the course: social interaction may lead during orientation, whereas individual cognitive activity may precede social peaks during resolution and consolidation.
RQ3. What temporal relationship exists between cognitive presence (access to cognitive resources) and social presence (forum participation) throughout the course?

5. Method

5.1. Experimental setting

The study was conducted within the framework of a training program for digital mentoring of older adults based on the ADDIE model (Analysis, Design, Development, Implementation, and Evaluation) [46, 47] (Fig. 1).

- Analysis: A review of scientific literature was conducted on cases of digital literacy practices through peer-to-peer methods, identifying the didactic gaps among mentors, given the diversity of the target group, which shares common characteristics such as medium-to-high cultural levels and intermediate or advanced digital skills. Additionally, the geographic dispersion of these individuals across the territory highlights the suitability of using digital platforms for communication and learning. Finally, a group of experts established the advisability of a modular and practical structure based on the main functions of digital mentoring practices.
- Design: A group of 10 experts developed each of the two modules: (a) digital mentoring, and (b) the design and development of digital literacy workshops. Each module was designed with reference to the CoI framework (Table 1).

- Development: For each module, the following materials were developed: (a) a didactic guide for instructors and learners, (b) interactive infographics and testimonial videos, (c) two videoconference sessions, (d) discussion forums on the module content, (e) assignment forum and (f) tasks based on case resolution and the design of a training workshop project. Each module was validated by a group of 40 experts in the field of communication and pedagogical innovation. To measure the quality of the materials, the "challenge" component of the environmental complexity model [48] was used, along with the subjective usability scale (SUS) [49].

Finally, a pilot test was conducted with a group of 14 older adults with intermediate or advanced digital skills, involving 6 instructors.

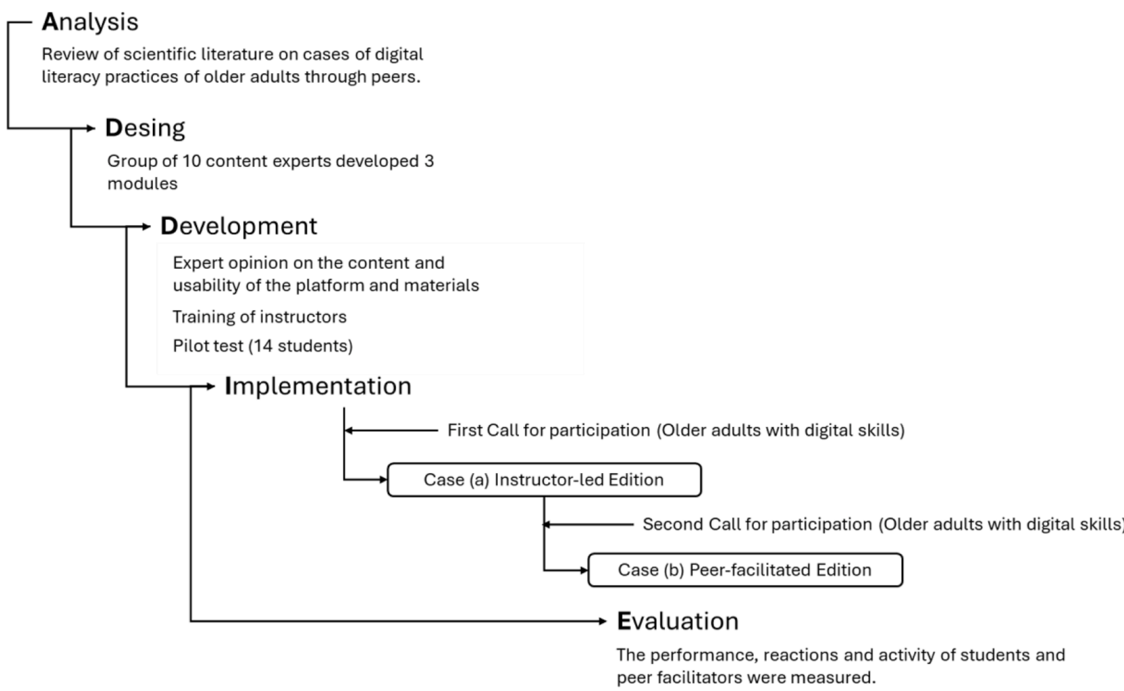

Fig. 1. ADDIE model applied to the intervention program.

Table 1
Design of the learning model based on the CoI framework.

Component	Purpose	Learning resources used
Teaching presence	To create a supportive learning climate, encourage interaction, and motivate participation. To provide feedback, synthesise ideas, and offer supplementary resources [20].	Videoconferencing sessions. Learning guides.
Social presence	To promote participation, mutual support, and social interaction among students [50].	Discussion forums on module topics Forums focused on each module's assessment assignment.
Cognitive presence	To support the construction of meaning and knowledge through deep thinking and collaboration [51].	Infographic materials covering module content. Interactive infographics explaining assignments. Discussion and assignment forums.

The two modules were implemented, and feedback was gathered from both the 14 participants and the 6 instructors.

- Implementation: Based on recommendations from the pilot test, the final version of the program simplified its content and tasks. Instructors were trained in the goals and structure of the final course version, its materials, and tasks. The instructors also participated in the course's development and pilot test. Participants were instructed on the program content and practiced using the e-learning platform. Furthermore, the platform was adapted to a user-friendly environment, including only the essential components for course access: (a) didactic guide, (b) interactive infographics, (c) discussion forums, and (d) assignment instructions. The course implementation used two editions: case (a), a traditional edition delivered by instructors where all students shared the same instructors; and case (b), an edition with peer facilitators, who were selected from among the students who had successfully completed the first edition (case (a)) and had expressed interest in participating in the second.

The role of the peer facilitators was to participate in the videoconference sessions and in the discussion forums. To ensure that the facilitators had the required skills, they were selected from among the graduate students of case (a). Clear expectations were set, and ongoing support was provided to the facilitators. Using peer facilitators in online courses for older adults also raised ethical considerations, so steps were taken to ensure that facilitators did not feel overburdened.

Both editions lasted two weeks, corresponding to the two modules. Edition (a) was implemented from Monday to Sunday, while edition (b) was implemented from Wednesday to Tuesday.

As aforesaid, this study placed special emphasis on ethical considerations. Informed consent was obtained from all participants, ensuring that they understood the study's objectives and that their participation was voluntary. Additionally, to protect participants' privacy, all collected and analysed data were anonymized. These measures ensure that the study adheres to ethical standards and respects the confidentiality of the information provided by the participants.

- Post-Course Evaluation: After the completion of both course versions, two components of the Kirkpatrick model were used to evaluate course quality: (a) Performance was assessed through Pre-Test and Post-Test evaluations. (b) Students' reactions were measured using a Likert-type scale, focusing on course structure, material usability, support received from instructors and peers, and perceived learning outcomes.

The performance test revealed significant improvements between the pre-test and post-test results. Additionally, the student reaction questionnaire yielded scores above 4 across all scales, based on a 1-to-5 rating system.

5.2. Participants

The participants were recruited through a call issued by the Government of Andalusia, Spain, which funds and coordinates social centres for older adults, as well as through the Red Cross and the State Association of University Programmes for Seniors. Additionally, to ensure that participants had an intermediate or advanced level of digital skills, a scale was applied.

Finally, 72 participants were selected for this study: 27 from the first edition (case (a)) and 45 from the second edition (case (b)). The mean age of the participants was 65.41 (SD = 6.51), 31 women and 41 men, and with 10 or more activity records in any of the Moodle events. A minimum threshold of 10 activity records per participant was set to ensure sufficient stability in the temporal correlation estimates and to avoid spurious patterns from very short trajectories. Nearby thresholds (8–12 records) were tested and produced comparable patterns, supporting the robustness of this criterion.

5.3. Operational mapping of log-based proxies to the constructs of the CoI framework

Following established practice in learning analytics research, this study operationalised selected behavioural indicators from Moodle logs as proxies for the three presences of the CoI framework. While trace data capture observable actions rather than latent psychological constructs, prior research has demonstrated that, when theoretically grounded and empirically validated, such indicators can meaningfully reflect cognitive, social, and teaching processes in online learning environments. This approach enables the temporal examination of CoI-related dynamics at a level of granularity that would be difficult to achieve through self-report or content-analysis methods alone

Teaching presence was operationalised through access to real-time videoconferencing. Synchronous sessions constitute the principal form of instructional scaffolding in many online courses; attendance logs can provide direct evidence of students' exposure to instructor guidance and clarification, therefore may serve as a proxy of teaching presence [52].

Social presence was operationalised via participation in assignment-related and thematic discussion forums. These behaviours represent opportunities for interpersonal interaction, collaboration, and affective communication, which are central to the conceptualisation of social presence within the CoI framework. Patterns of communication and collaboration in e-learning environments can be analysed to infer social and participatory engagement [11].

Cognitive presence was operationalised through learners' interactions with instructional resources, including accesses to content pages, learning guides, assignment descriptions, and assignment submissions. Although these behaviours do not directly capture internal cognitive states, a substantial body of research supports their use as valid proxies for cognitive engagement when grounded in theory and validated empirically. Studies have shown that specific clickstream behaviours—such as viewing guides or consulting content pages—correlate with self-reported cognitive engagement and with coded indicators of cognitive presence in discussion transcripts [11,23,52].

In particular, accessing learning guides has been repeatedly associated with the early phases of meaning-making described in the CoI framework. Baldwin and Ching [41] and Martin et al. [42] report that learners frequently consult guides when orienting themselves to tasks, clarifying expectations, or preparing to integrate new information—behaviours aligned with the triggering and exploration phases of cognitive presence. Recent studies linking LMS trace data with cognitive-presence coding further show that resource-access patterns often precede or accompany deeper cognitive engagement [15,17].

Taken together, this evidence supports the theoretical and empirical plausibility of treating these behavioural traces as partial but meaningful indicators of teaching, social, and cognitive presence.

To ensure replicability, we established explicit coding rules for transforming raw Moodle logs into CoI-related indicators (Fig. 2). Only events generated by enrolled participants were included. System-generated events, instructor-only actions, and duplicate entries created by page refreshes were removed. Each event was assigned to a single CoI construct based on its pedagogical function.

Each log entry contained the following fields: timestamp, user ID, component/ and event name.

The coding procedure consisted of four steps:

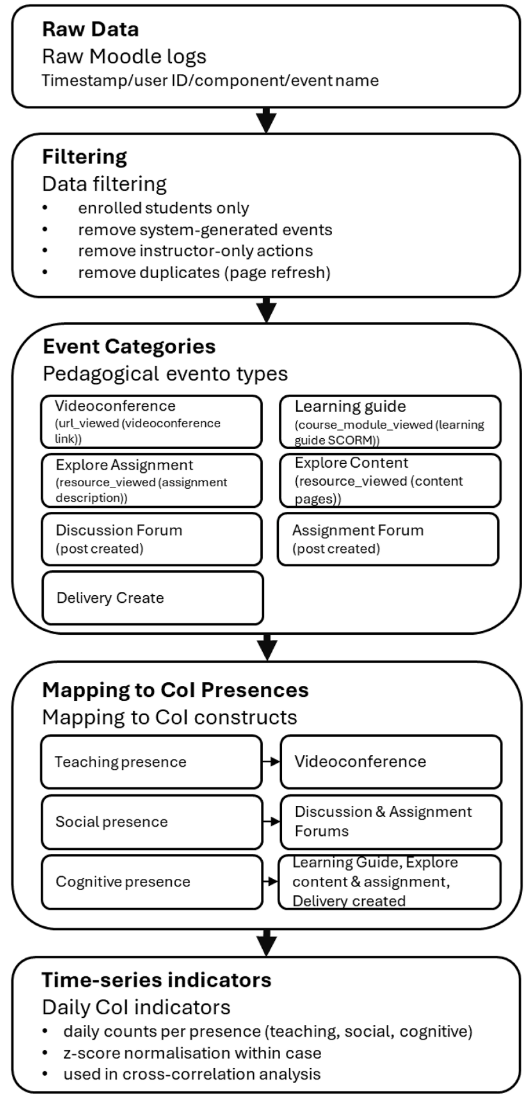

Fig. 2. Visual summary of the log-based proxy mapping to CoI constructs.

(1) Event selection and filtering. Only events generated by enrolled participants were retained. System-generated events, instructor-only actions, and duplicated entries created by page refreshes were removed. Events were included only if they corresponded to one of the predefined pedagogical components of the course (videoconferences, learning guides, content pages, assignment descriptions, assignment submissions, and discussion forums).
(2) Mapping of Moodle events to CoI constructs. Each event type was assigned to a CoI presence based on its pedagogical function.
(3) Temporal aggregation. All events were aggregated at the day × presence level. For each day of the 14-day course, we computed the total number of events per CoI category. This produced three parallel time series per case (teaching, social, cognitive presence), each with 14 observations.
(4) Normalisation and handling of missing data. To allow comparison across cases with different enrolment sizes, each time series was normalised using z-scores (mean = 0, SD = 1) within each case. Days with no recorded activity for a given presence were coded as zero prior to normalisation. No interpolation was applied; instead, missing days were treated as structurally meaningful inactivity.

In line with the aim of the study, 4276 records were selected

Table 2
Mapping of Moodle log events to CoI constructs.

CoI Construct	Event categories	Moodle Event Code	Event Description	Operational Rule Applied	Case (a) (n = 27)	Case (b) (n = 45)
Teaching Presence	Videoconference	url_viewed (videoconference link)	Student clicks an external URL to access the synchronous videoconference session	Counted as one instance of teaching presence; only events linked to scheduled synchronous sessions were included	69	217
Social Presence	Discussion Forum	post_created (Discussion Forum)	Student creates a new forum post	Counted as social presence; instructor posts excluded	126	237
	Assignment Forum	post_created (Assignment Forum)	Student creates a new forum post	Counted as social presence; instructor posts excluded	317	147
Cognitive Presence	Learning Guide	course_module_viewed (learning guide SCORM)	Student views the learning guide	Treated as activation of planning/ triggering phase of cognitive presence	184	300
	Explore Content	resource_viewed (content pages)	Student views content pages or infographics	Counted as cognitive presence; only module-related resources included	249	371
	Explore Assignment	resource_viewed (assignment description)	Student views assignment instructions	Counted as cognitive presence; treated as task-orientation behaviour	684	1264
	Delivery Created	delivery created	Student uploads a file or begins completing the submission form, and clicks "submit for grading"	Counted as cognitive presence; interpreted as evidence of task engagement	45	66
Total					1674	2602

(Table 2).

5.4. Data analysis

To identify interaction patterns among log-based proxies of CoI-related activity, the records were arranged in chronological order using SPSS software, version 27.0.1.0.

The variables analysed were the time series corresponding to each of the seven units of analysis over a 14-day period, which matched the duration of the course. To address RQ1, the variables used were access to videoconferences, access to the learning guide, exploration of content, exploration of assignments, and participation in forums. To address RQ2, the variables included access to the learning guide, exploration of assignments, and participation in both discussion forums and assignment forums. For RQ3, the variables comprised access to the learning guide, exploration of content, exploration of assignments, and participation in both types of forums.

Prior to commencing the analysis, (a) all metrics were subjected to a normalisation procedure (mean = 0, standard deviation = 1), ensuring comparability across variables and preventing those with wider numerical ranges from disproportionately influencing the results; (b) records were aggregated into uniform temporal intervals and organised into analytically coherent categories, which facilitated the construction of comparable series with sufficient density for robust examination; and (c) unequal levels of participation were mitigated by applying minimum activity thresholds, thereby ensuring that the correlations captured patterns representative of the dataset as a whole rather than being skewed by extreme cases.

To examine how the variables were related over time, the following steps were undertaken: (a) each variable was transformed into a time series; (b) a Pearson correlation analysis was conducted to explore relationships among variables; and (c) a cross-correlation analysis was performed to assess the time lag between two series (e.g. how a change in one variable may affect another after a certain period).

Cross-correlation is a technique that quantifies the extent to which a series of activities is correlated with others at different time points, helping understand how they influence one another over time. In Moodle, cross-correlations were used to identify relationships between variables, both when they occurred simultaneously and when delayed effects were present. For the 14 observations, following the recommendations of Shumway & Stoffer [53], the analysis was restricted to ± 3 lags, which allowed for a minimally reasonable number of overlapping pairs (ideally ≥ 10).

Given the short length of the time series (14 daily observations per case), the stability and interpretability of cross-correlation estimates must be approached with caution. Short time series tend to increase sampling variability, reduce the precision of lagged coefficients, and limit the statistical power to detect temporal associations. For this reason, the cross-correlation analysis conducted in this study is framed as exploratory, aimed at identifying indicative temporal patterns rather than providing confirmatory statistical evidence.

To enhance transparency, 95% confidence intervals were computed for all cross-correlation coefficients. These intervals quantify the uncertainty associated with each estimate and are reported alongside the coefficients. Correlations falling near conventional significance thresholds are interpreted cautiously.

Because the analysis involves multiple cross-correlations across variables and lags, we considered the issue of multiple comparisons. However, given the short time series and the exploratory aims of the study, applying conservative corrections (e.g., Bonferroni) would substantially inflate Type II error rates and obscure potentially meaningful temporal patterns. Instead, we emphasise effect sizes, confidence intervals, and theoretical coherence when interpreting the results.

In total, approximately **294 cross-correlations** were computed (147 per case), resulting from the 21 possible combinations among the seven variables and the seven lags considered (lag 0, ± 1, ± 2, ± 3). This volume of comparisons contextualises the decision not to apply formal multiple-comparison corrections within an exploratory analytical framework.

Finally, all variables were constructed from daily aggregated cohort-level click counts, meaning that the resulting temporal patterns reflect macro-level course rhythms rather than individual learning trajectories. This distinction is highlighted in the Discussion to avoid over-generalisation of the findings.

6. Results

6.1. Case (a)

6.1.1. Descriptive results

Table 3 shows a significant progression in student participation across the various course activities. The "learning guide" variable displays notable peaks at the beginning of each module (38 in Module 1 and 30 in Module 2. "Videoconference" also features prominently at the start of each module. Participation in "discussion forum" and "assignment forum" is moderate yet consistent. The variables "explore assignment" and "explore content" show sustained growth, reaching peaks of 127 and 69 respectively, which suggests active engagement with the learning resources. Finally, "delivery created" remains low overall, with occasional spikes.

Table 3
Count and percentage of log-based proxies of CoI-related activity (case (a)).

	Date	Video conference	Learning Guide	Explore Assignment	Explore Content	Discussion Forum	Assignment Forum	Delivery Created
Mod.1	10.06.24	21	38	39	36	7	51	0
	11.06.24	0	11	16	19	4	13	0
	12.06.24	0	11	14	11	6	8	0
	13.06.24	0	12	10	14	8	20	0
	14.06.24	0	4	51	4	4	23	6
	15.06.24	0	6	42	8	7	5	7
	16.06.24	0	7	69	11	3	14	8
Mod.2	17.06.24	36	30	127	29	7	32	8
	18.06.24	0	8	36	16	3	28	0
	19.06.24	0	9	59	18	19	22	2
	20.06.24	0	5	38	18	6	16	4
	21.06.24	1	4	27	24	8	26	0
	22.06.24	0	4	30	12	1	12	3
	23.06.24	0	5	36	5	0	7	6

6.1.2. Results of the correlation analysis and cross-correlation

6.1.2.1. Correlation analysis. A Pearson correlation analysis was conducted using the normalised values of each series. Table 4 presents the correlations among the recorded log-based proxies of CoI-related activity.

Table 4 reveals significant relationships among various educational activities. "Videoconferencing" is strongly associated with the use of the "learning guide" ($r = 0.870$, $p < .01$), as well as with "assignment exploration" ($r = .720$, $p < .01$) and "content exploration" ($r = .711$, $p < .01$), suggesting that students who participate in synchronous sessions also actively engage with the available learning resources.

> "Videoconferencing" is also moderately linked to the "assignment forum" ($r = .649$, $p < .05$), although other correlations with the "discussion forum" are weak and statistically non-significant ($r = .100$ to $.344$, $p > .05$).
> "Learning guide" shows a high correlation with "content exploration" ($r = .795$, $p < .01$) and with the "assignment forum" ($r = .760$, $p < .01$). Likewise, "content exploration" is significantly related to the "assignment forum" ($r = .809$, $p < .01$).

Finally, the "delivery created" variable presents negative correlations with several resources, such as the "learning guide" ($r = -0.256$, $p > .05$) and "content exploration" ($r = -0.485$, $p > .05$), without statistical significance.

6.1.2.2. Cross-correlation analysis. The cross-correlation analysis shows that certain activities are strongly related to each other at different time lags. Table 5 presents the data within the range of −3 to 2, as no significant correlations were found at other time lags.

The cross-correlation analysis of activity logs in Moodle reveals significant relationships between various activities with time lags. Below is a summary of the key findings (Fig. 3).

Table 4
Pearson correlations between time series of log-based proxies of CoI-related activity (case (a)).

	1	2	3	4	5	6	7
1. Videoconference	1						
2. Learning Guide	.870**	1					
3. Explore Assignment	.720**	.399	1				
4. Explore Content	.711**	.795**	.276	1			
5. Discussion Forum	.100	.181	.159	.344	1		
6. Assignment Forum	.649*	.760**	.300	.809**	.259	1	
7. Delivery Created	.022	-.256	.517	-.485	-.191	-.381	1

*p<.05, **p<.01.

Regarding RQ1, the results show positive and highly significant correlations between the use of "videoconferencing" and other key activities. The relationship with the "learning guide" is particularly strong ($r = .867$, $p = .000$), suggesting that students who participate in videoconferencing sessions also engage intensively with the learning guides. Significant correlations are also observed with "explore assignment" ($r = .720$, $p = .003$), "explore content" ($r = .711$, $p = .002$), and the "assignment forum" ($r = .649$, $p = .010$), all with an optimal lag of 0, indicating that these activities occur simultaneously. The correlation with the "discussion forum" ($r = .732$, $p = .002$) appears with a lag of 2, reflecting forum participation two days after accessing the videoconference.
Regarding RQ2, the "learning guide" variable is strongly related to "explore content" ($r = .795$, $p = .000$) and "assignment forum" ($r = .760$, $p = .000$), both with a lag of 0, suggesting that students consult these resources concurrently. The correlation with "discussion forum" ($r = .572$, $p = .042$, lag 2) is also significant, indicating possible interaction two days later. In contrast, the relationship with "delivery created" is negative ($r = -.491$, $p = .085$), although not statistically significant, which may suggest that those who consult the guides do not typically submit work immediately. Correlations between "explore assignments" and other variables are weaker, with significance observed only in relation to the "discussion forum" ($r = .642$, $p = .013$, lag 2), suggesting that students tend to participate in discussion forums two days after accessing the assignment.
Regarding RQ3, a very strong correlation is observed between "explore content" and "assignment forum" ($r = .809$, $p = .000$), indicating that students who explore content also actively participate in submission-related forums. In contrast, the relationship with "discussion forum" is weak and not statistically significant ($r = .344$, $p = .165$), suggesting that content exploration is not directly linked to participation in open discussions.

6.2. Case (b)

6.2.1. Descriptive results

Table 6 shows a clear progression in student participation across the two course modules. The "learning guide" variable reflects a high initial level of interest, with 57 records on the first day of Module 1 and 49 at the start of Module 2. "Videoconferencing" also features prominently at the beginning of each module. Participation in "discussion forum" and "assignment forum" remains moderate yet consistent, with peaks of up to 31 posts in the discussion forum and 102 in the assignment forum. The "explore assignment" and "explore content" variables show progressive growth, reaching peaks of 143 and 30 respectively. Finally, "delivery created" remains relatively low, although with occasional spikes that coincide with assignment deadlines.

Table 5
Cross-correlation between log-based proxies of CoI-related activity (case (a)).

	Series A	Series B	Optimal Lag	r (SE)	95% CI	p-value
RQ1	Videoconference	Explore Content	0	.711 (.267)	[.266, .899]	0.002
	Videoconference	Learning Guide	0	.867 (.267)	[.595, .958]	0.000
	Videoconference	Explore Assignments	0	.720 (.267)	[.281, .903]	0.003
	Videoconference	Assignment Forum	0	.649 (.267)	[.165, .873]	0.010
	Videoconference	Discussion Forum	2	.732 (.289)	[.292, .907]	0.002
RQ2	Learning Guide	Explore Content	0	.795 (.267)	[.450, .929]	0.000
	Learning Guide	Delivery Created	1	-.491 (.277)	[–.822, .145]	0.085
	Learning Guide	Explore Assignments	0	.399 (.267)	[–.207, .792]	0.318
	Learning Guide	Discussion Forum	2	.572 (.289)	[.037, .855]	0.042
	Learning Guide	Assignment Forum	0	.760 (.267)	[.389, .915]	0.000
	Explore Assignments	Explore Content	–3	–529 (.302)	[–.846, .104]	0.073
	Explore Assignments	Delivery Created	0	.517 (.267)	[–.018, .836]	0.077
	Explore Assignments	Discussion Forum	2	.642 (.289)	[.154, .870]	0.013
	Explore Assignments	Assignment Forum	–2	.378 (.289)	[–.235, .780]	0.352
RQ3	Explore Content	Discussion Forum	0	.344 (.267)	[–.270, .764]	0.165
	Explore Content	Assignment Forum	0	.809 (.267)	[.476, .934]	0.000

Note: Coefficients correspond to cross-correlations at the identified optimal lag. The 95% confidence intervals reflect the uncertainty associated with each estimate; relationships with wide intervals should be interpreted cautiously due to the short length of the time series.

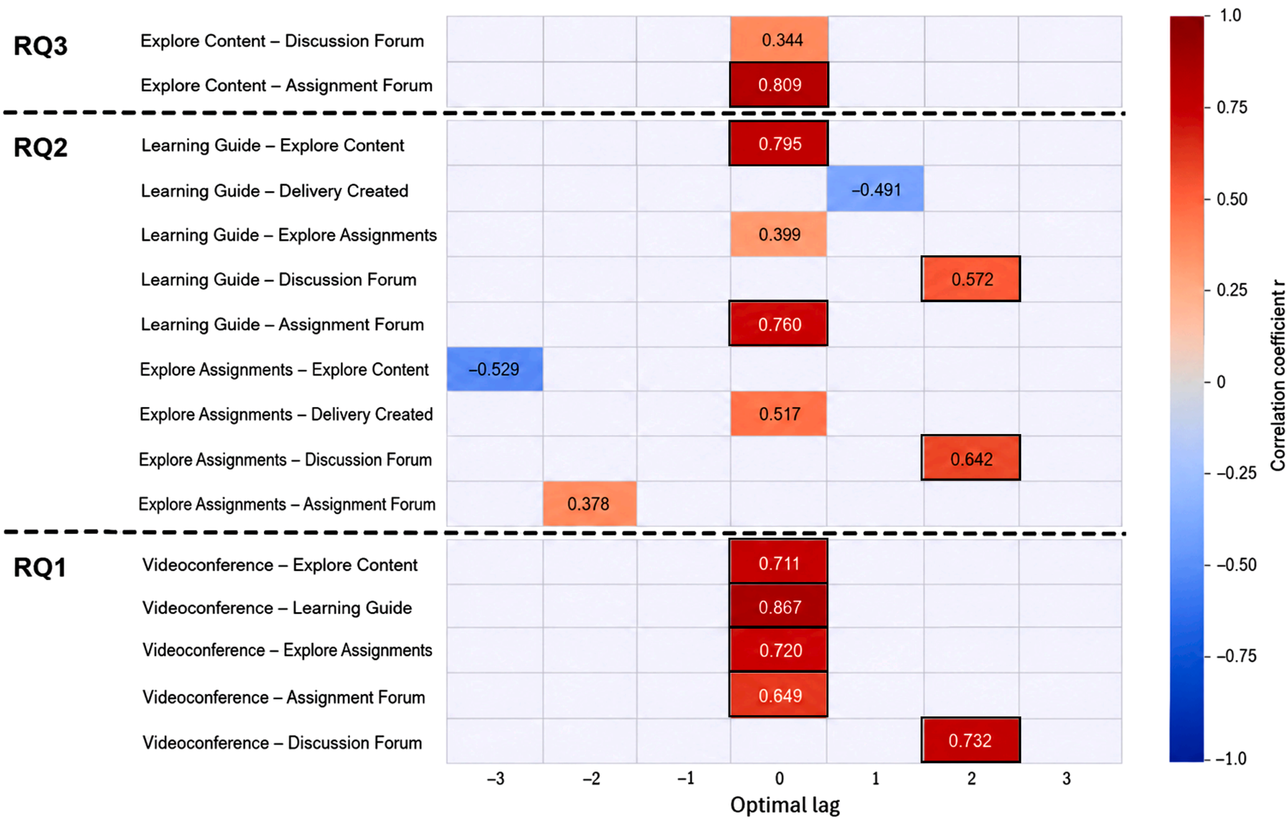

Fig. 3. Diagram with cross-correlations between log-based proxies of CoI-related activity (case (a)).
Note: Significant correlations are highlighted with a bolded box.

6.2.2. Results of the correlational analysis and cross-correlation

6.2.2.1. Correlations analysis. Table 7 reveals significant relationships among various course activities, shedding light on student behavioural patterns. "Videoconferencing" shows a very strong correlation with the "learning guide" ($r = .858$, $p < .01$) and "explore content" ($r = .762$, $p < .01$). In turn, the "learning guide" is strongly associated with "explore content" ($r = .891$, $p < .01$).

The "explore assignment" variable displays a strong correlation with "delivery created" ($r = .805$, $p < .01$), reflecting that task exploration is closely linked to the act of submitting work. In contrast, "assignment forum" shows weak or negative correlations with most variables, which may suggest a more independent or less integrated use within the overall activity flow.

Finally, "discussion forum" is moderately associated with the "learning guide" ($r = .505$) and "explore content" ($r = .764$, $p < .01$) variables, indicating that participation in discussions is related to engagement with learning resources.

6.2.2.2. Cross-correlation analysis. Table 8 and Fig. 4 presents the correlations between variables at different time lags. Time lags ranging from -3 to 3 were used.

Table 6
Count and percentage of log-based proxies of CoI-related activity (case (b)).

	Date	Videoconference	Learning Guide	Explore Assignment	Explore Content	Discussion Forum	Assignment Forum	Delivery Created
Mod. 1	06.11.24	56	57	94	84	11	102	2
	07.11.24	8	6	84	7	9	51	7
	08.11.24	0	3	29	4	3	16	3
	09.11.24	0	7	24	6	8	18	1
	10.11.24	0	9	115	22	20	97	11
	11.11.24	0	13	128	10	31	96	5
	12.11.24	0	32	94	25	5	44	10
Mod. 2	13.11.24	72	49	80	42	0	26	1
	14.11.24	0	17	44	12	0	12	0
	15.11.24	0	11	31	17	9	43	0
	16.11.24	0	6	18	6	2	25	0
	17.11.24	0	16	57	18	3	47	2
	18.11.24	0	12	143	22	5	95	14
	19.11.24	0	19	125	30	5	54	10

Table 7
Pearson correlations between time series of log-based proxies of CoI-related activity (case (b)).

	1	2	3	4	5	6	7
1. Videoconference	1						
2. Learning Guide	.858**	1					
3. Explore Assignment	.104	.223	1				
4. Explore Content	.762**	.891**	.357	1			
5. Discussion Forum	.312	.505	.208	.764**	1		
6. Assignment Forum	-.153	-.121	.468	-.008	.092	1	
7. Delivery Create	-.256	-.072	.805**	.083	-.024	.157	1

**p<.01.

Table 8
Cross-correlation between log-based proxies of CoI-related activity (case (b)).

	Series A	Series B	Optimal Lag	r (SE)	95% CI	p-value
RQ1	Videoconference	Explore Content	0	.762 (.269)	[.336, .923]	0.015
	Videoconference	Delivery Created	–1	.354 (.277)	[–.270, .764]	0.225
	Videoconference	Learning Guide	0	.858 (.267)	[.584, .957]	0.007
	Videoconference	Explore Assignments	3	-.543 (.302)	[–.846, .106]	0.097
	Videoconference	Assignment Forum	–2	.628 (.289)	[.139, .866]	0.050
	Videoconference	Discussion Forum	0	.312 (.267)	[–.296, .748]	0.265
RQ2	Learning Guide	Explore Content	0	-.891 (.267)	[–.967, –.623]	0.005
	Learning Guide	Delivery Created	3	-.529 (.302)	[–.841, .120]	0.105
	Learning Guide	Explore Assignments	3	-.742 (.302)	[–.917, –.260]	0.030
	Learning Guide	Discussion Forum	0	.505 (.267)	[–.031, .833]	0.082
	Learning Guide	Assignment Forum	–2	.594 (.289)	[.093, .853]	0.062
	Explore Assignments	Explore Content	–3	-.595 (.302)	[–.868, .045]	0.072
	Explore Assignments	Delivery Created	0	.805 (.267)	[.457, .932]	0.010
	Explore Assignments	Discussion Forum	–2	-.322 (.289)	[–.780, .292]	0.287
	Explore Assignments	Assignment Forum	0	.468 (.267)	[–.141, .812]	0.105
RQ3	Explore Content	Discussion Forum	0	.764 (.267)	[.341, .924]	0.014
	Explore Content	Assignment Forum	2	.288 (.289)	[–.325, .741]	0.338

Note: Coefficients correspond to cross-correlations at the identified optimal lag. The 95% confidence intervals revealed substantial uncertainty around several coefficients, particularly those with non-zero lags.

Regarding RQ1, the results show positive and significant correlations between the "videoconference" variable and several key resources. Its strong association with the "learning guide" (r = .858, p = .007) and "explore content" (r = .762, p = .015), both with an optimal lag of 0, suggests that students who participate in synchronous sessions also actively consult study materials. A further significant correlation is observed with the "assignment forum" (r = .628, p = .050, lag –2), indicating that "forum participation" may precede "videoconferencing" by two days. In contrast, the relationship with "explore assignments" is negative (r = –.543, p = .097, lag 3), although not statistically significant, which may reflect a temporal disconnect between these activities. Correlations with "delivery created" and "discussion forum" are weak and not significant.

Regarding RQ2, the "learning guide" variable shows a very strong negative correlation with "explore content" (r = –.891, p = .005). Negative correlations are also observed with "explore assignments" (r = –.742, p = .030, lag 3) and "delivery created" (r = –.529, p = .105), although the latter is not statistically significant. In contrast, a positive correlation is detected with the "assignment forum" (r = .594, p = .062, lag –2) and the "discussion forum" (r = .505, p = .082), indicating that the use of guides may be linked to participation in collaborative spaces. Additionally, the "explore assignments" variable is positively associated with "delivery created" (r = .805, p = .010).

Regarding RQ3, a significant correlation is observed between "explore content" and the "discussion forum" (r = .764, p = .014), indicating that students who explore content also actively engage in discussions. Moreover, although the relationship with the "assignment forum" is weak and not significant (r = .288, p = .338, lag 2), it

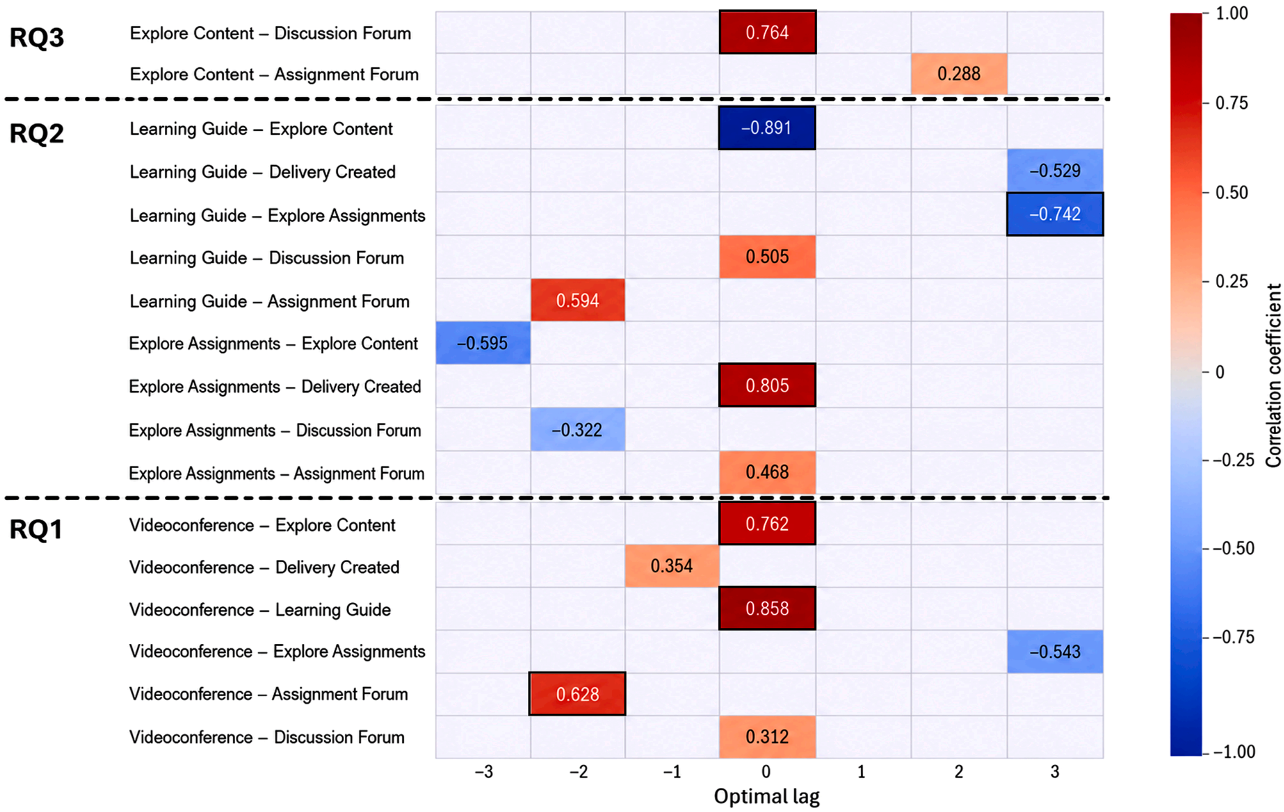

Fig. 4. Diagram with cross-correlations between log-based proxies of CoI-related activity (case (b)).
Note: Significant correlations are highlighted with a bolded box.

suggests that "content exploration" may precede participation in task-related forums.

7. Discussion

The results obtained in both cases (a) and (b) reveal temporal patterns that reflect the interaction between log-based proxies of CoI-related activity [51]. However, given that behavioural traces capture observable actions rather than latent cognitive or social processes, the interpretations offered here should be understood as exploratory and indicative rather than confirmatory

(a) Regarding RQ1

In both cases, and at the same time point, videoconferencing appears as a relevant element associated with cognitive engagement. Videoconferencing has been linked to the fostering of social presence [54] and teaching presence, as instructors guide discussions and support learning during these sessions. In both case studies, these sessions were observed in connection with cognitive activities such as engaging with the learning guide and exploring course content, indicating a temporal association between videoconferencing and proxies of cognitive activity. This pattern is consistent with research emphasising the role of synchronous teaching presence in shaping asynchronous learning processes [54,55].

Nevertheless, some of the lag-0 correlations observed may reflect the instructional scheduling of the course (i.e., timing of synchronous sessions) rather than learner-driven temporal coupling. This distinction is important because it limits the extent to which synchronous peaks can be interpreted as evidence of cognitive activation triggered by teaching presence.

Recent studies suggest that videoconferencing may enhance content comprehension and encourage participation in subsequent activities, such as forum discussions [55]. The delayed correlation observed between videoconferencing and forum participation in case (a) could be interpreted in relation to cognitive overload [56], which occurs when individuals receive more information than they can process, thereby affecting their problem-solving and decision-making abilities. In contrast, in case (b), the non-significant but simultaneous correlation points to a possible mediating role played by peer facilitators, who may have contributed to alleviating the cognitive load during videoconferencing sessions [50]. However, these interpretations remain tentative, as the study design does not allow us to isolate the specific mechanisms underlying these temporal associations.

(b) Regarding RQ2

The correlation between access to content, learning guides, and forum participation suggests moments of cognitive activation, although this interpretation must be tempered by the temporal patterns observed and by the limitations inherent in aggregated data. In case (a), the delayed correlation between task exploration and participation in discussion forums points to a sequence in which learners process information before engaging in interaction, a pattern consistent with the integration phase of cognitive presence described by Garrison et al. [51]. However, some lag-0 correlations may reflect the instructional scheduling of the course—such as the timing of synchronous sessions—rather than learner-driven temporal coupling, which limits the extent to which synchronous peaks can be interpreted as direct evidence of cognitive activation.

The findings also support the view that the learning guide functions as an orienting resource that helps learners navigate materials and engage with the various proposed activities [57]. Recent studies indicate that access to well-designed guides enables students to plan and organise their actions autonomously, fostering cognitive presence from the outset [41] and increasing the likelihood of participation in collaborative tasks

[42]. In case (a), access to the learning guide was associated not only with engagement with course content, but also with participation in task forums at the same time point, and with a two-day lag in relation to participation in the discussion forum.

In contrast, in case (b), a strong—though not statistically significant and accompanied by a wide confidence interval—correlation was observed at the same time point between access to the guide and participation in the discussion forum. This pattern should be interpreted cautiously, as it may reflect alternation between resource types, differences in task sequencing, or cognitive load effects rather than a direct or inverse form of cognitive engagement. Moreover, because click counts were aggregated at cohort level, the temporal patterns identified represent macro-level rhythms of the course rather than individual learning trajectories, limiting the extent to which these findings can be interpreted as evidence of individual cognitive or social processes.

(c) Regarding RQ3

Cognitive activities such as content and task exploration, and social activities such as participation in discussion forums and collaborative tasks, exhibit temporal relationships that vary according to context. In both cases (a) and (b), content exploration correlates with posts in both discussion and task forums when no time lag is applied. In case (a), a two-day lag shows that exploration of the learning task precedes posts in discussion forums, illustrating how cognitive presence (thinking and reflecting) and social presence (interacting with others) complement one another over short time intervals. Learners appear to alternate between processing content and engaging with peers, either synchronously or with delay. These alternating patterns observed between individual content exploration and socially oriented participation are consistent with findings from MOOC research, where cognitive engagement has been shown to emerge through temporally structured contributions in discussion forums [26]. This convergence suggests that short-term engagement cycles may reflect general cognitive processes rather than course-specific artefact. Similarly, studies have highlighted that learners tend to move back and forth between solitary and collaborative activities to deepen their understanding, which can positively influence academic performance [58].

These temporal fluctuations can also be interpreted through Zimmerman's [34] self-regulated learning model, in which learners adjust their levels of interaction and reflection in response to task demands. However, because SRL was not directly measured in this study, any connection to self-regulatory processes must be treated as hypothetical and requires further empirical validation. Even so, the pattern observed here is consistent with research showing that social presence does not necessarily precede or follow cognitive presence in a linear manner; instead, both presences mutually reinforce one another in cycles of participation [24,43].

In case (b), the non-significant but simultaneous correlation between access to the learning guide and forum participation suggests an immediate form of social activation, potentially shaped by contextual features of the course. Although causality cannot be inferred, prior work indicates that course design and the presence of peer facilitators can modulate the intensity and timing of social presence by synchronising it with individual cognitive actions [21]. The convergence between our findings and those of Liu et al. [26] suggests that such short-term engagement cycles may reflect general cognitive processes rather than course-specific artefacts.

Finally, given that participants were older adults, it is important to consider how age-related factors—such as digital literacy, familiarity with the platform, and the mentoring training context—may have shaped the observed temporal patterns. Older adults may engage in more linear or sequential interaction patterns, rely more heavily on guides, or require additional time to transition between tasks, all of which could influence the rhythms captured in the time-series analysis.

(d) Strengths and contributions of the study

This study makes several contributions to research on online learning and the Community of Inquiry framework. Methodologically, it introduces a temporal analytic approach—combining time-series modelling and cross-correlation analysis with transparent proxy operationalisation—that offers a dynamic view of online engagement rarely examined in CoI studies. This strengthens the interpretability and replicability of log-based analyses.

The comparative design, involving an instructor-led and a peer-facilitated course edition, provides insight into how different facilitation structures may shape short-term engagement rhythms, without aiming to compare their effectiveness. This adds nuance to existing work on the role of instructional and peer support in online environments.

The study also contributes to the literature on older adult learners. Although it does not compare age groups, the temporal patterns observed align with prior research on ageing and digital learning, suggesting more sequential engagement, stronger reliance on structured guidance, and sensitivity to cognitive load. Examining these behaviours through a temporal lens offers a novel perspective on how older adults navigate online learning.

Finally, the temporal analyses offer theoretical insights into the CoI framework. The lagged patterns observed suggest that elements of teaching presence—particularly synchronous guidance—tend to precede subsequent increases in cognitive presence, indicating a temporally delayed rather than immediate influence. These dynamics also point to a sequential coupling among presences that static analyses cannot capture, highlighting how instructional scaffolding may initiate short-term cycles of exploration and interaction.

8. Practical implications for instructional design

The temporal associations identified in this study offer several design considerations for digital learning environments. Although these insights are based on correlations and do not imply causality, they may inform future practice: (a) use videoconferencing as a potential point of cognitive activation, followed by asynchronous activities that encourage individual reflection and information processing; (b) develop learning guides that connect content with collaborative tasks, supporting both learner autonomy and opportunities for social interaction; and (c) monitor the temporal rhythms of learner activity to adjust course pacing and promote a balanced alternation between individual and collaborative work, in line with self-regulated learning principles. These considerations reflect patterns observed in a small, context-specific sample and should be validated in broader and more diverse settings.

9. Limitations and future research

Several limitations should be considered when interpreting the findings. First, the study relies on log-based behavioural proxies, which capture observable actions rather than latent constructs. Although theoretically grounded, these indicators offer only partial representations of teaching, social, and cognitive presence, introducing potential threats to construct validity.

Second, while self-regulated learning is incorporated into the theoretical framework, it was not directly measured; any interpretation linking temporal patterns to self-regulatory processes is therefore speculative.

Tirth, the short duration of the course (14 days) also constrains the robustness of the time-series analysis, as the limited number of observations reduces the stability of cross-correlation estimates and increases the likelihood that some patterns reflect short-term fluctuations.

Finally, some of the temporal associations identified—particularly those at lag 0—may reflect the structure and scheduling of the course (such as the timing of synchronous sessions) rather than learner-driven temporal coupling. This limits the extent to which the findings can be interpreted as evidence of endogenous learning rhythms.

Future research should combine temporal log-data analysis with direct measures of self-regulated learning, qualitative data, or discourse analysis to strengthen construct validity. It would also be valuable to extend the duration of courses to obtain more stable time series. Further studies could compare different age groups or facilitation models to examine how temporal engagement patterns vary according to learner characteristics and instructional design.

10. Conclusions

This study shows that log-based activity proxies in Moodle, aligned with the CoI framework, can reveal meaningful temporal patterns of interaction. These traces offer insight into the real-time dynamics of cognitive and social presence, while also requiring caution, as they capture observable behaviour rather than underlying cognitive or social processes.

The results highlight significant associations among proxies such as content access, learning guides, forum participation, and videoconferencing, contributing to a clearer understanding of how interactions unfold in virtual environments and supporting the methodological potential of the CoI framework. However, these patterns should be interpreted as exploratory rather than causal, given the short time series and the influence of instructional scheduling.

The study advances CoI research by emphasising the temporal dimension of teaching, social, and cognitive presence, suggesting that these presences may fluctuate in coordinated or lagged cycles. The inclusion of older adult learners further underscores the need to consider digital literacy, technological familiarity, and contextual factors when interpreting behavioural data in online learning settings.

Funding

This research was funded by the following projects:

- Grant TED2021-129253B-100 funded by MICIU/AEI/10.13039/501100011033 and by the European Union “NextGenerationEU/PRTR”.
- Grant PID2021-123552OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by “ERDF/EU”.
- Project PROYEXCEL_00320 funded by the Department of Universities, Research and Innovation of the Junta de Andalucía (PAIDI 2020).

Ethics statement

The study was approved by the Ethics Committee of Andalusia (SICEIA-2025–001,665), Spain, on 31 October 2025.

Declaration of AI use

During the preparation of this manuscript, the authors used Microsoft Copilot for proofreading and language refinement. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the final version of the manuscript.

Ethics approval information

The study adhered strictly to the fundamental ethical principles governing research involving human participants. All individuals were informed about the objectives of the study, its voluntary nature, and their right to withdraw at any time without consequence. Written informed consent was obtained, and all data were pseudonymised prior to analysis using Moodle’s native functionality, in line with ethical standards that safeguard confidentiality and privacy. These procedures complied with the ethical provisions of the Declaration of Helsinki and the national regulations applicable to educational research.

The dataset was reviewed by the host institution and subsequently published on Zenodo; the DOI has been withheld to preserve anonymity during peer review.

CRediT authorship contribution statement

Ramón Tirado-Morueta: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Rocío Illanes-Segura:** Writing – review & editing, Validation, Resources, Methodology, Investigation. **Juan Carlos Infante-Moro:** Writing – review & editing, Resources, Methodology, Investigation. **Ana M. Duarte-Hueros:** Writing – review & editing, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ramon Tirado-Morueta reports article publishing charges and writing assistance were provided by Spain Ministry of Science Innovation and Universities. Ramon Tirado Morueta reports financial support, equipment, drugs, or supplies, and travel were provided by Department of Universities, Research and Innovation of the Junta de Andalucía. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge the support received from the Spanish State Research Agency, the Ministry of Science, Innovation, and Universities, the European Union “NextGenerationEU/PRTR,” and “ERDF/EU” [Grant TED2021-129253B-100, Grant PID2021-123552OB-I00]. They also thank the Regional Ministry of Universities, Research, and Innovation of the Government of Andalusia [Project PROYEXCEL_00320]. The authors express their gratitude to all participants and to the organizations that facilitated contact with the facilitators and moderators: the Senior Citizens Department of the Regional Ministry of Social Inclusion, Youth, Families, and Equality of the Government of Andalusia; the Click-A program of the Andalusian Red Cross; and the Aula de la Experiencia at the University of Huelva.

Data availability

The dataset supporting this article is openly available at: Tirado-Morueta, R., Duarte-Hueros, A., & Illanes-Segura, R. (2025). *Dataset on Cross-Correlation Patterns between Activity Records in Moodle Based on the Community of Inquiry (CoI) Model* [Data set]. Zenodo. htt ps://d oi.org/10.5281/zenodo.17551830

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